

Glaucoma

1. Overview

- Optic neuropathy → progressive peripheral vision loss
- Elevated IOP is the **only modifiable risk factor**
- Most types are **chronic and asymptomatic** until advanced

Type	Key Features	Pharmacist Role
Open-Angle (OAG) — most common	Chronic, asymptomatic early. Normal or high IOP. Late: peripheral vision loss.	Counsel on adherence, technique, ADEs. Screen meds for drug-induced causes.
Acute Angle-Closure — EMERGENCY	Eye pain, blurred vision, halos, headache, N/V. Sudden IOP spike.	REFER to ER / ophthalmologist immediately.
Chronic Angle-Closure	Similar to OAG, angle narrowed, often asymptomatic.	Same as OAG. Counsel re: anticholinergic drug risk.

2. When to Refer

- **URGENT (same day):** acute angle-closure — eye pain, halos, blurred vision, N/V, headache → ER or ophthalmologist
- **Non-urgent:** new/suspected diagnosis, medication failure, intolerance, worsening symptoms, family history with no recent exam
- **Corticosteroid users:** recommend ophthalmic exam if patient has additional glaucoma risk factors

3. Risk Factors

Drug-Induced (screen every patient's medication list)

- **Corticosteroids** (any route: inhaled, intranasal, ophthalmic, systemic, topical) — most common drug cause
- **Anticholinergic drugs** (antidepressants, antihistamines, antispasmodics, decongestants) — can precipitate acute angle-closure in predisposed patients
- **Topiramate** — associated with acute angle-closure
- **OCP >3 years** — may be associated with higher prevalence of primary OAG

Patient Risk Factors

- **OAG:** age >60, family history, myopia, type 2 diabetes, hypertension, migraine, elevated IOP
- **Angle-closure:** hyperopia, female sex, advanced age, family/personal history
- **Note:** up to 50% of glaucoma patients have normal IOP (<21 mmHg) — normal IOP does not rule out glaucoma

4. Goals of Therapy

- Lower IOP to an individualized target — adjusted if progression continues at target
- Prevent/halt progressive visual field loss and optic nerve damage
- Relieve pain and improve vision (acute forms)
- Minimize medication burden and preserve quality of life

5. Treatment Algorithm — Open-Angle Glaucoma

Step up if: optic disc cupping progresses, visual field deteriorates, or IOP inadequately controlled.

Step	Treatment
1st	Topical prostaglandin analogue (preferred — most effective, once daily, fewest systemic ADEs) OR topical beta-blocker OR laser trabeculoplasty (preferred if: elderly, adherence issues, pseudoexfoliation, pigment dispersion)
2nd	Add or substitute: prostaglandin analogue OR beta-blocker OR topical CAI OR alpha-2 agonist OR laser trabeculoplasty
3rd	Add agents (usually up to 3 topical agents). Consider fixed-combination products to improve adherence. Prostaglandin + beta-blocker + CAI or alpha-2 agonist or cholinergic agonist (pilocarpine) OR laser trabeculoplasty
4th	Laser trabeculoplasty ± topical medications ± oral CAI (acetazolamide)
5th	Glaucoma surgery (trabeculectomy, drainage tube) ± medications ± laser

- **Acute angle-closure:** definitive = laser iridotomy / surgical iridectomy. Aggressive medical treatment (topical beta-blocker, alpha-2 agonist, pilocarpine, IV acetazolamide) used as a bridge until procedure.

6. Drug Classes

Prostaglandin Analogues — 1st Line

Mechanism: increase aqueous outflow (uveoscleral ± trabecular pathway)

Drug	Dose	Mechanism	Key ADEs	Contraindications / Notes
Latanoprost 0.005% (Xalatan, generics) <\$15	1 drop HS	Uveoscleral outflow ↑	Conjunctival hyperemia (least vs others), iris darkening, eyelash growth, stinging	Most cost-effective. Do not exceed once daily (more frequent → less effective). Refrigerate unopened; stable 6 wk at room temp after opening.
Bimatoprost 0.01%, 0.03% (Lumigan RC, Vistitan) \$45–60	1 drop HS	Uveoscleral + trabecular ↑	More hyperemia than latanoprost. Iris darkening, eyelash ↑	Do not exceed once daily.
Travoprost 0.003%, 0.004% (Izba, Travatan Z) \$15–30	1 drop HS	Uveoscleral outflow ↑	Hyperemia, iris darkening, eyelash ↑	Non-BAK preservative (ionic buffer) — preferred if BAK sensitivity.
Latanoprostene bunod (Vyzulta) \$15–30	1 drop HS	Uveoscleral + trabecular ↑	Similar to other prostaglandins	Do not exceed once daily.

Beta-Blockers — 1st Line Option

Mechanism: decrease aqueous humor production

Drug	Dose	Mechanism	Key ADEs	Contraindications / Notes
Timolol 0.25%, 0.5% (Timoptic, generics) <\$15 Gel (Timoptic XE) <\$15	Drops: 1 drop QAM or Q12H Gel: 1 drop once daily	Aqueous production ↓ (non-selective β-blocker)	Bronchospasm, bradycardia, hypotension, CHF exacerbation, depression, masking hypoglycemia, impotence	CI: asthma, severe COPD, bradycardia, AV block, decompensated CHF. Gel: shake once before use.
Betaxolol 0.25% (Betoptic S) <\$15	1 drop QAM or Q12H	Aqueous production ↓ (β1-selective)	Less bronchospasm than timolol. Similar cardiac effects.	May use with caution in mild COPD (β1-selective). Still avoid in asthma/severe COPD.

Alpha-2 Adrenergic Agonists

Mechanism: decrease aqueous production + increase uveoscleral outflow

Drug	Dose	Mechanism	Key ADEs	Contraindications / Notes
Brimonidine 0.2% (Alphagan, generics) <\$15	1 drop Q12H	Aqueous ↓ + uveoscleral ↑ (alpha-2 selective)	Ocular allergy (most common), dry mouth, fatigue, hypotension, headache	CI: MAO inhibitors. Not for infants (apnea). Contains BAK.
Brimonidine 0.15% (Alphagan P, generics) <\$15	1 drop Q8H	Same as above	Slightly lower ocular allergy rate than 0.2%	Contains purite (non-BAK) — preferred if BAK sensitivity. CI: MAOIs.
Apraclonidine 0.5%, 1% (Iopidine) \$15–30	1–2 drops Q8–12H	Same as above	>40% incidence of ocular allergy/blepharoconjunctivitis	NOT for chronic use. 1% used perioperatively. CI: MAOIs.

Topical Carbonic Anhydrase Inhibitors (CAIs)

Mechanism: inhibit aqueous humor production. Good alternative if beta-blocker contraindicated.

Drug	Dose	Mechanism	Key ADEs	Contraindications / Notes
Dorzolamide 2% (Trusopt, generics) <\$15 PF: Trusopt PF \$45–60)	Monotherapy: 1 drop Q8H Adjunct: 1 drop Q12H	Aqueous production ↓	Bitter/sour taste, stinging, local allergy	PF formulation available for BAK allergy (align pipette fins side-to-side to eye corners).
Brinzolamide 1% (Azopt) \$15–30	1 drop Q12H (can increase to Q8H after 4 wk if needed)	Aqueous production ↓	Bitter taste, stinging (slightly better tolerated than dorzolamide)	Shake well before use.

- **Sulfa allergy and CAIs:** cross-reactivity with sulfa antibiotics is extremely low — **NOT contraindicated**. Monitor when initiating if history of severe reactions (anaphylaxis, SJS).

Cholinergic Agonists (Pilocarpine) — Later Line

Mechanism: muscarinic stimulation → ciliary muscle contraction → increases trabecular outflow

Drug	Dose	Mechanism	Key ADEs	Contraindications / Notes
Pilocarpine 1%, 2%, 4%(Isopto Carpine) <\$15	1 drop QID	Trabecular outflow ↑	Miosis (reduced night vision), accommodative spasm/myopia, brow ache, GI upset (rare), retinal detachment (rare)	Poorly tolerated in children and young adults. QID dosing → poor adherence. 4th-line agent.

Oral Carbonic Anhydrase Inhibitors — Emergencies / Refractory Only

Reserved for acute angle-closure emergencies or cases refractory to all topical therapy. **~50% cannot tolerate long-term.**

Drug	Dose	Mechanism	Key ADEs	Contraindications / Notes
Acetazolamide(generics) <\$15	250–1000 mg/day PO in divided doses	Aqueous production ↓	Paresthesias, metabolic acidosis, hypokalemia, GI upset, urolithiasis, fatigue, depression. Rare: aplastic anemia, SJS.	CI: hepatic dysfunction, CrCl <10 mL/min. Monitor electrolytes.
Methazolamide(generics) \$45–60	50–100 mg BID–TID	Aqueous production ↓	Milder GI/CNS effects than acetazolamide	CI: hepatic dysfunction, renal insufficiency.

7. Fixed-Combination Products (Canada)

Benefits: improve adherence, reduce preservative exposure, prevent drop washout

Product	Brand	Dose	Notes
Brimonidine 0.2% / Timolol 0.5%	Combigan	1 drop Q12H	CI: asthma, COPD, MAOIs.
Brinzolamide 1% / Timolol 0.5%	Azarga	1 drop Q12H (max Q8H)	CI: asthma, COPD. Shake well.
Dorzolamide 2% / Timolol 0.5%	Cosopt (PF available)	1 drop Q12H	CI: asthma, COPD. PF option for BAK allergy.
Latanoprost 0.005% / Timolol 0.5%	Xalacom, generics	1 drop once daily	CI: asthma, COPD. Do not exceed once daily.
Travoprost 0.004% / Timolol 0.5%	DuoTrav PQ, generics	1 drop once daily	CI: asthma, COPD. Non-BAK preservative.
Brimonidine 0.2% / Brinzolamide 1%	Simbrinza	1 drop Q12H (max Q8H)	Beta-blocker free — use if respiratory/cardiac CI to timolol. CI: MAOIs. Contains BAK.

8. Counselling

Eye Drop Technique

- **Nasolacrimal occlusion:** press inner corner of eye for 2–3 min after instillation — reduces systemic absorption (especially critical with beta-blockers)
- **Alternatively:** keep eyelids closed for 3–5 minutes after instillation
- **Multiple drops:** wait at least 5 minutes between different eye drops (prevents washout)
- **Soft contact lenses:** remove before instilling drops; reinsert after 15 minutes
- **Prostaglandins:** apply in the evening. Do not use more than once daily — more frequent dosing reduces efficacy

Preservatives

- **BAK (benzalkonium chloride):** most common preservative. Up to 6% of glaucoma patients allergic. Prolonged exposure can damage corneal surface (irritation, dryness, burning).
- **Non-BAK alternatives:** Travoprost/Travatan Z (ionic buffer), Brimonidine 0.15%/Alphagan P (purite), Trusopt PF, Cosopt PF, DuoTrav PQ

OTC / Drug Interactions

- **OTC antihistamines and decongestants:** intermittent use is usually safe in open-angle glaucoma. In angle-closure — anticholinergics can precipitate acute attack in predisposed patients (elderly, hyperopic females, no iridectomy).
- **MAO inhibitors:** contraindicated with brimonidine and apraclonidine
- **Systemic beta-blockers:** additive effect with topical timolol — monitor heart rate and blood pressure
- **Cannabis:** NOT recommended (Canadian Ophthalmological Society). Short duration, psychotropic effects, and CBD may elevate IOP and interfere with THC's IOP-lowering effect.

9. Monitoring

Parameter	When	Action
IOP (by ophthalmologist/optometrist)	Each follow-up visit	Adjust target downward if progression continues
Visual field and optic disc	Periodically (ophthalmologist)	Refer if deteriorating
Adherence to eye drops	Each refill	Counsel on technique; consider fixed-combo if adherence poor
Ocular ADEs (hyperemia, allergy, irritation)	Ongoing	Switch to PF formulation or different class
Systemic ADEs with beta-blockers (HR, BP, dyspnea)	Especially on initiation	Withhold and refer if bradycardia or bronchospasm
Electrolytes / renal function (oral CAs)	During use	Discontinue if metabolic acidosis or hypokalemia
Corticosteroid users (any route)	Assess for risk factors at each encounter	Recommend ophthalmic exam if additional risk factors present

10. Drug Choice by Patient Profile

Patient Situation	Preferred / Avoid
Default 1st line	Topical prostaglandin analogue (latanoprost is most cost-effective, generics <\$15)
Asthma or severe COPD	AVOID all topical beta-blockers and all timolol-containing combos. Use PGA + topical CAI or brimonidine. Simbrinza (brimonidine/brinzolamide) is beta-blocker free.
Mild COPD (not severe asthma)	Betaxolol (β 1-selective) may be used with caution
On MAO inhibitors	AVOID brimonidine and apraclonidine
BAK preservative sensitivity	Travoprost (Travatan Z), Brimonidine 0.15% (Alphagan P), Trusopt PF, Cosopt PF, DuoTrav PQ
Poor adherence / multiple drops	Fixed-combination products. Consider laser trabeculoplasty.
Elderly with adherence/cost issues	Laser trabeculoplasty (first-line option for this group)
Sulfa allergy on CAI therapy	NOT contraindicated. Monitor on initiation, especially if history of severe reaction (anaphylaxis, SJS).
Cannabis inquiry	Do not recommend. Short duration, psychotropic effects, CBD may elevate IOP.